Lab Assignment-7.1

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Task Description #1 (Syntax Errors — Missing Parentheses in Print Statement) Task:

Provide a Python snippet with a missing parenthesis in a print statement (e.g., print "Hello"). Use AI to detect and fix the syntax error.

Requirements:

Run the given code to observe the error.

Apply AI suggestions to correct the syntax.

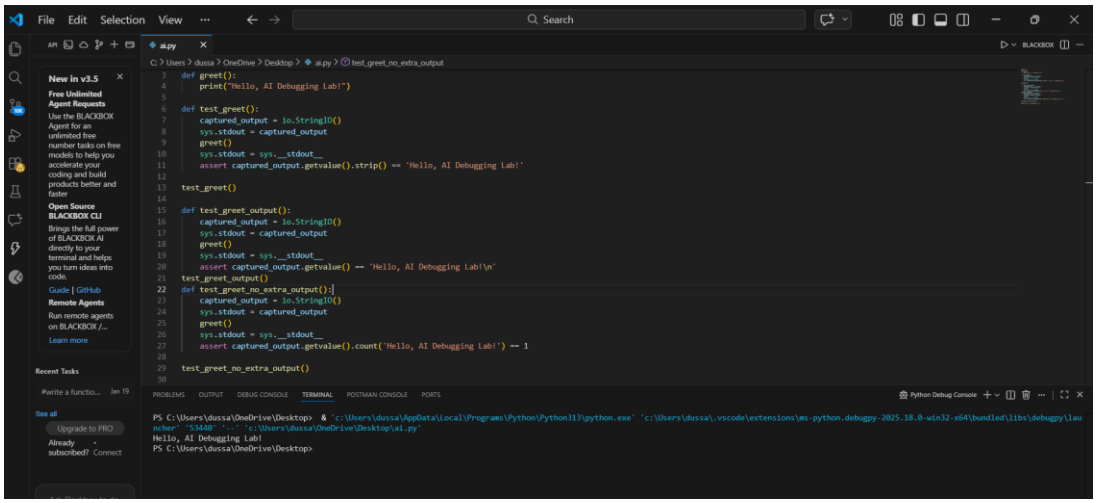
Expected Output #1 :

- Corrected code with proper syntax and AI explanation.

Prompt:

#Detect and fix the syntax error in this Python code:

#def greet(): print 'Hello, AI Debugging Lab!' ; greet() #Explain the issue, provide corrected Python 3 code, and include 3 assert tests to verify correct output.



Explanation:

The issue in the original code is that it uses Python 2 syntax for the print statement. In Python 3, print is a function and requires parentheses.

Task Description #2 (Incorrect condition in an If Statement)

Task:

Supply a function where an if-condition mistakenly uses = instead of Let Al identify and fix the issue.

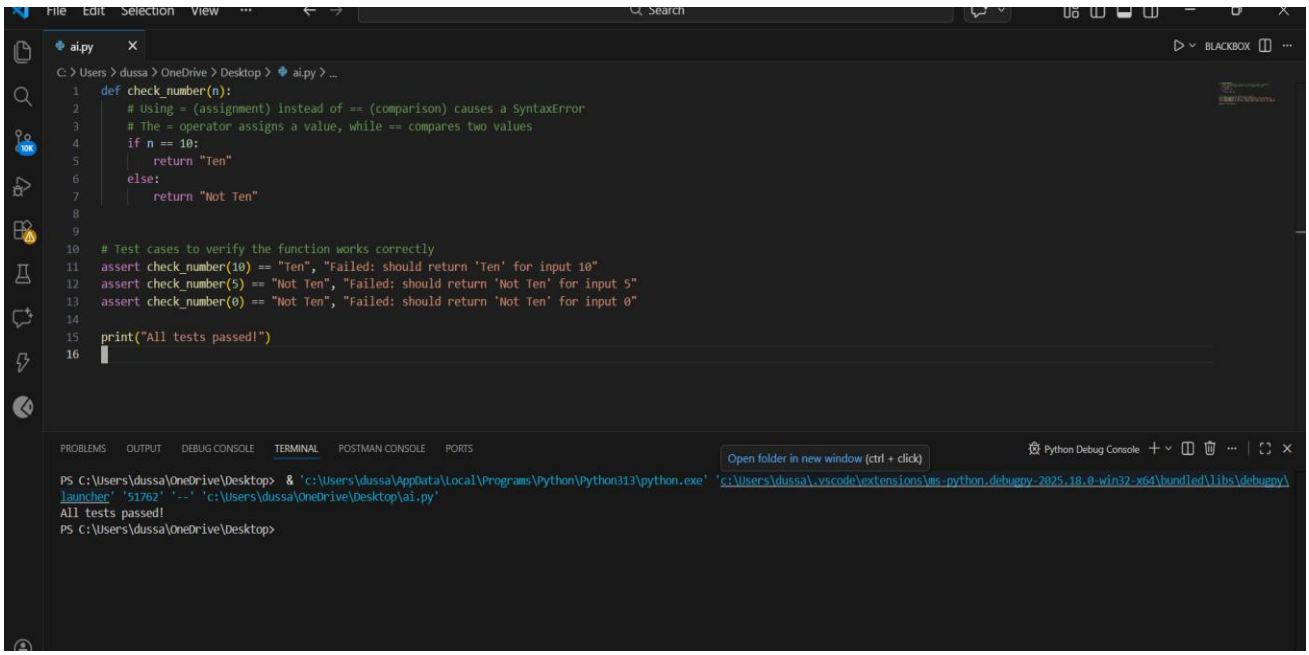
```
# Bug: Using assignment (=) instead of comparison (==)
    def check_number(n):
        if n = 10:
            return "Ten"
        else:
```

s.

Prompt used:

Detect and fix the error in this Python code: def check_number(n): if n = 10: return "Ten" else: return "Not Ten". #Explain why using causes a bug, provide corrected code using '==',and include 3 assert tests to verify the function works.

Code:



Explaination :

The error occurs because = is an assignment operator, not a comparison operator. In an if condition, Python requires == to compare values, and using = causes a SyntaxError. Replacing = with == correctly checks whether n is equal to 10, allowing the function to work properly.

Task Description #3 (Runtime Error — File Not Found)

Task:

Provide code that attempts to open a non-existent file and crashes. Use Al to apply safe error handling.

```
# Bug: Program crashes if file is missing
def read_file(filename):
with open(filename, 'r') as f:
return f.read()
print(read_file("nonexistent.txt"))
```

Requirements:

• Implement a try-except block suggested by AI.
• Add a user-friendly error message.
• Test with at least 3 scenarios: file exists, file missing, invalid Path.

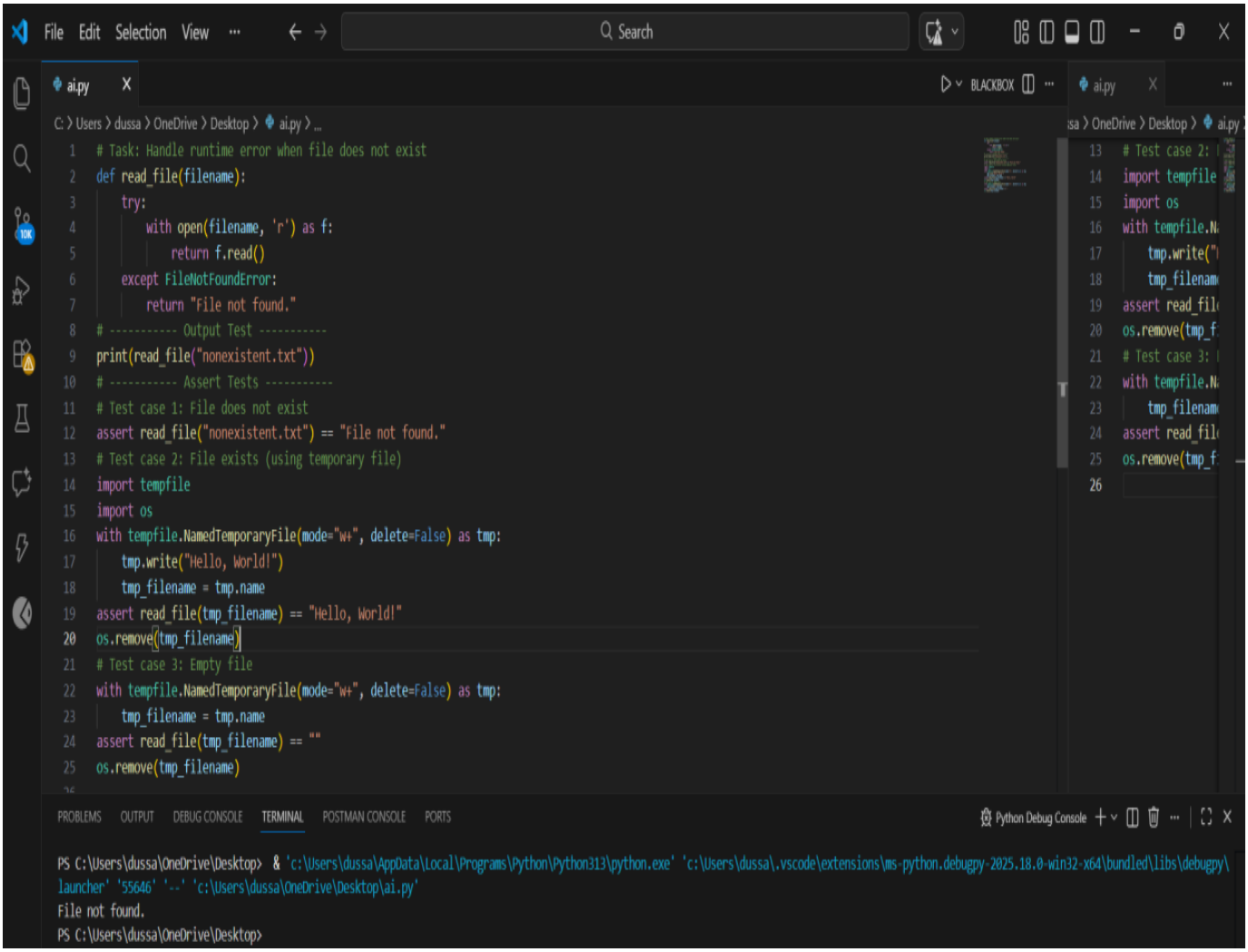
Expected Output #3:

- Safe file handling with exception management.

Prompt:

#Analyze this Python code that crashes when a file does not exist: def read_file(filename): with open(filename,'r') as f: return f.read()
#Fix the runtime error using try—except, add a user-friendly error message, and include 3 assert tests for file exists, file missing, and invalid path cases.

code:



Explanation:

The program crashes because it tries to open a file that does not exist, which raises a FileNotFoundError. By using a try—except block, the exception is caught and handled safely instead of stopping the program. This allows the function to return a userfriendly error message while continuing execution.

TASK:

Give a class where a non-existent method is called (e.g.,

obj.undefined_method()). Use Al to debug and fix.

```
# Bug: Calling an undefined
method class Car:
def start(self):
return "Car started"
 my_car = Car()
print(my_car.drive())  #
drive() is not defined
```

Requirements:

- Students must analyze whether to define the missing method or correct the method call.
- Use 3 assert tests to confirm the corrected class works.

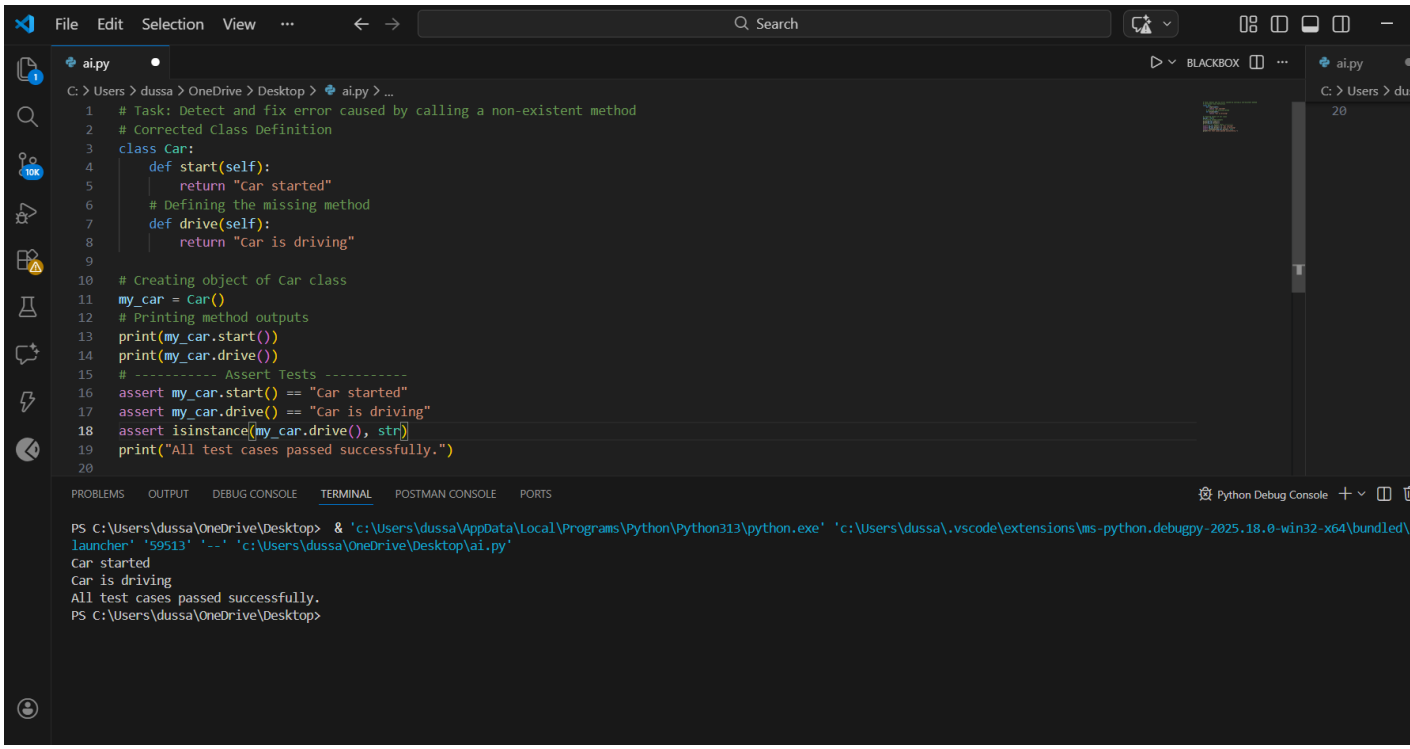
Expected Output #4:

- Corrected class with clear Al explanation.

Prompt used:

#Detect and fix the error in this Python code where a nonexistent method is called: class Car: def start(self): return "Car started" my_car = Car(); print(my_car.drive()) #Explain the cause of the error, decide whether to define the missing method or correct the method call, provide corrected code, and include 3 assert tests to verify the class works properly.

Code :



Explaination:

The error occurs because the method drive() is called on the Car object, but it is not defined in the Car class, which raises an

Task Description #5 (TypeError — Mixing Strings and Integers in Addition)

Task:

Provide code that adds an integer and string 5" + 2) causing

a TypeError. Use Al to resolve the bug.

Bug: TypeError due to mixing string

and integer def add_five(value): return

value + 5 print(add_five("

10"))

Requirements:

• Ask Al for two solutions: type casting and string concatenation.

• Validate with 3 assert test cases.

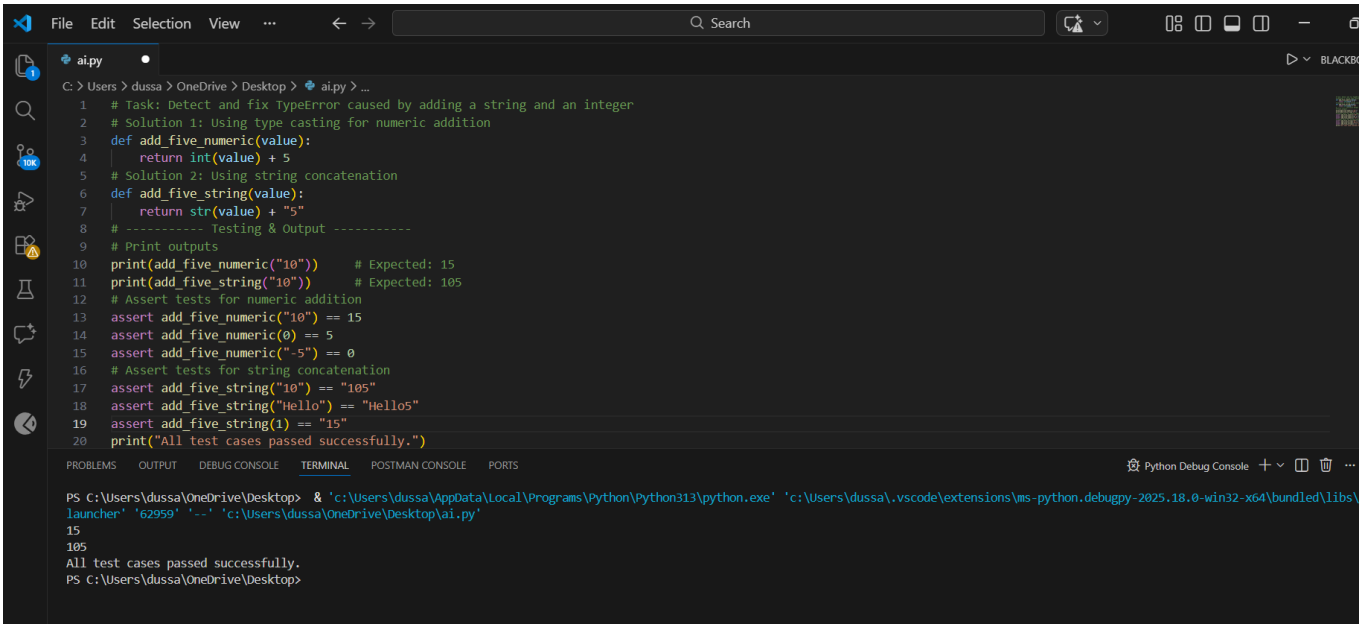
Expected Output #5:

• Corrected code that runs successfully for multiple inputs.

Prompt used:

#Detect and fix the TypeError in this Python code caused by adding a string and an integer: def add five(value): return value + 5 ; print(add_five("10"))#Explain why the error occurs, provide two solutions (one using type casting for numeric addition and one using string concatenation), and include 3 assert tests to verify both solutions work correctly.

Code:



Explaination:

The error occurs because Python cannot add a string and an integer, which raises a TypeError. One solution is to convert the input to an integer using type casting so numeric addition can be performed. Another solution is to convert the value to a string and use string concatenation instead, depending on the intended behavior.